

Long Live the Patient King ^{*}

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Abstract

I propose a simple model of the relationship between economic growth and extractive taxation. A citizen and a ruler interact over time. The citizen owns a stock of capital which, in every period, yields some output. The ruler chooses how much output to expropriate for his private consumption, subject to the citizen's participation constraint. The growth rate of output is endogenous to the citizen's choice of effort. In this context, I define a *social contract* as a (Subgame Perfect) equilibrium of the repeated game. Moreover, a social contract is *extractive* at a given history if it leaves the citizen indifferent between the on path continuation and the outside option; and it is *non-extractive* otherwise.

The key tension of the model is that the citizen's effort in a given period increases output only in the future, when the ruler can always expropriate it. It follows that when players are myopic, any social contract is extractive. By contrast, when players are forward looking, any Pareto efficient and renegotiation proof social contract features a *dual transition*. As time passes, society endogenously moves from a regime of slow growth — where extractive social contracts are implementable — to a regime of fast growth — where they are not.

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1 Introduction

As famously argued by Weingast [1995], "the fundamental political dilemma of an economic system" is that "a government strong enough to protect property rights and enforce contracts is also strong enough to confiscate the wealth of its citizens," thereby deterring investments and growth (North and Weingast [1989]).

However, history shows that the dilemma can be resolved. Over the very long run, the world has become both more prosperous and less despotically governed, often in a correlated manner (Figure 1).

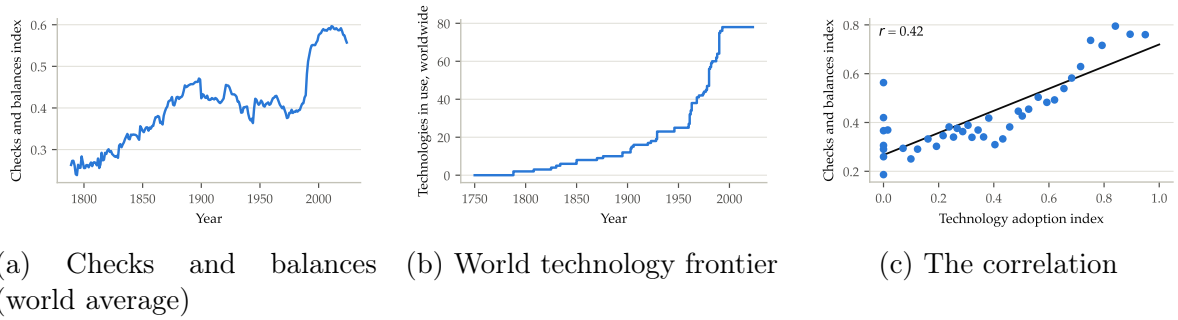


Figure 1: **Constraints on rulers and technological progress in the long run.**

Note: See Appendix C for details.

North and Weingast [1989] provides a possible solution to this puzzle that emphasizes the role played by constitutions and formal institutions more in general. In this view, constitutions are devices that force rulers to commit ex ante to non-extractive plans they would otherwise renegotiate ex post. A related and influential strand of literature describes these commitments as the "fundamental" cause of growth and human progress (e.g., Acemoglu et al. [2001, 2005]).

Yet, a society's successful transition out of an extractive regime might signal a deeper change in the strategic interaction between citizens and rulers, of which the signing of a new constitution could be just a symbol or a symptom. Moreover, constitutions and formal systems of checks and balances must often be self-enforcing and robust to players' incentives to renegotiate them.

This article describes how societies can escape extraction and achieve sustained growth when neither rulers nor citizens have commitment power, and when institutional change must be robust to renegotiation.

I build a simple model where a citizen (she) and a ruler (he) repeatedly interact over an infinite horizon. The citizen owns the means of production. She possesses a stock of capital which in every period produces some output according to a time-varying technology. The ruler has political power. He can tax the citizen's output at his discretion and freely consume the collected tax revenues.

The stage game is as follows. At the beginning of each period, the ruler announces a tax rate. The citizen observes it, and chooses one of three possible actions. She either quits the game, stays in the game and exerts (costly) effort, or stays in the game saving effort. If the citizen quits the game, she secures a fixed outside option, and the ruler's payoff is zero forever thereafter. Otherwise, the game continues. In this case, her choice of effort determines the per-period growth rate of the economy (fast if effort is exerted, slow otherwise). The ruler maximizes life-time consumption. The citizen maximizes life-time consumption net of the cost of effort.

In this model, the economy grows at a fast rate in every period (a situation I refer to as "sustained growth") if and only if two conditions are satisfied. The first condition is immediate: the citizen must not quit the game. In this sense, the existence of a citizen-ruler *continued* relationship (which I then define as a *social contract*) is a necessary (but not sufficient) condition for growth. This feature of the model captures the idea that if the citizen quits the game, she loses access to her capital because — in the absence of a ruler — property rights are not enforced. The second condition is that the citizen exerts effort in every period.

The main obstacle, however, is that any return to the citizen's effort in period t is realized only in period $t + 1$, when the ruler can always expropriate it by increasing the tax rate. Anticipating this risk, the citizen may have no incentives to exert effort — exactly as noted by [North and Weingast \[1989\]](#) and [Weingast \[1995\]](#).

When this happens in equilibrium, the social contract is *extractive*. The ruler raises taxes in every period up to the point in which the citizen's participation constraint binds. In turn, she never exerts effort and growth is minimized. Crucially, when players are myopic, this is the unique possible equilibrium at every history.

Relatedly, any *non-extractive* social contract (i.e., an equilibrium where the citizen enjoys a positive surplus) must be sustained by non-Markovian strategies, whereby players' strategies are functions of the history of the game. In this sense, non-extractive social contracts appear as *relational contracts* (e.g., [Levin \[2003\]](#)), that is, agreements sustained by the fact that players value the future dividends of their relationship.

My main result is that when players are forward looking, sustained growth and low rent extraction are not only possible but also inevitable (up to to a simple refinement on the equilibrium concept). On the path of *every* Pareto efficient and renegotiation proof Subgame Perfect equilibrium (or social contract), the simple society described in my model overcomes the "fundamental dilemma" of [Weingast \[1995\]](#) by means of a system of repeated and self-enforcing promises. In every period, the ruler promises low taxes in the next period in exchange for the citizen's effort. Crucially, in the next period, the ruler has an incentive to keep his promise as he wants to induce further growth. Put simply, the faster growth of the pie compensates the ruler for his reduced share.

However, the parient ruler must live long enough. The citizen may have to wait some periods before the fast growth path starts to be in the ruler's interest. At the beginning of the game, extractive social contracts are implementable, and the citizen does never exert effort. Then, after a certain calendar period is reached, the political-economic equilibrium changes: the citizen starts exerting effort, and the ruler always reduces the tax rate accordingly. From that period onwards, every social contract is *non-extractive* as it must leave the citizen with a positive surplus.

On the path of these equilibria, the citizen's surplus is necessarily larger than the cost of effort, so that her net consumption flow strictly increases as a consequence of the ruler's tax concession. This is a consequence of the dynamic nature of the model. The citizen advances the cost of effort one period before being repaid for it. Each period of effort can then be seen as a one-period loan from the citizen to the ruler, so that its repayment must cover the principal (the cost of effort) *and* the interest, which the citizen captures as a profit. For this reason, the citizen is necessarily a strict beneficiary of sustained growth, and not merely reimbursed for starting it.

The transformation that occurs in equilibrium has both political and economic consequences. A famous parallel is the *dual revolution* of Hobsbawm [1962], which describes the emergence of the modern world as a joint consequence of the French Revolution and of the English industrial revolution. I call this equilibrium property of the model a *dual transition* to emphasize Hobsbawm's idea that institutional and economic change are not parallel events, but two halves of a single process.

In this model, the key engine of the dual transition is the accumulation of technology, or knowledge. The returns to the citizen's effort are increasing in the level of productivity already attained (a *standing on shoulders* property).¹ Therefore, as time passes and productivity (slowly) increases, these returns are higher. It is then increasingly valuable for the ruler to reduce rent extraction and concede a non-extractive social contract in order to induce the citizen's effort and achieve further growth.

The historical reading of this result is that modern rulers are less extractive than ancient ones not because constitutions or other formal institutions constrain them, but because citizens' work has become so productive that discouraging it through extraction is no longer in the rulers' own interest. In this framework, the concession of constitutions reflects a deeper transformation of the ruler's own incentives, which is ultimately triggered by technological progress.

The inevitability of non-extractive social contracts shows that players can be successfully deterred from unilateral, "selfish" deviations. The ruler can be deterred from raising the tax rate once the citizen has already exerted effort, and the citizen can be deterred from saving the cost of effort once the ruler has reduced rent extraction. To deter such deviations, social contracts prescribe punishments.

By subgame perfection, punishments must be itself equilibria of the off path continuation. By the adopted notion of renegotiation proofness, such equilibria must be Pareto undominated. That is, following a deviation, at the beginning of every subsequent period, there must exist no other equilibrium which would make both players strictly better off. The intuition behind this refinement is that if such alternative Pareto dominating equilibrium existed, no player would oppose a renegotiation of the punishment.

¹The name refers to Isaac Newton's famous aphorism: "If I have seen further, it is by standing on the shoulders of giants."

A direct consequence of renegotiation proofness is that off path continuations cannot prescribe a permanently slower growth path whenever faster growth would be both socially efficient and incentive compatible. Unlike classic work on folk theorems, players' "cooperation" can then be sustained in equilibrium for any discount factor.

To further illustrate the off path properties of non-extractive social contracts, I focus on two particular contracts: the citizen- and ruler-preferred contracts. These two contracts correspond to the two boundaries of the set of renegotiation proof social contracts (which corresponds to the Pareto frontier). Under the ruler-preferred (respectively, citizen-preferred) contract, the citizen's (respectively, ruler's) value function is at its lower bound at every history. Off path, the two contracts serve as each other's punishment. A deviation by the ruler triggers a one-period pause of effort followed by the citizen-preferred contract, and a deviation by the citizen triggers the ruler-preferred one. The harshest credible punishment is thus not to permanently shirk the pie, but to hand the entire surplus of the relationship over to the victim of the deviation.

The remainder of the paper is organized as follows. In the next paragraph, I briefly summarize the related literature. Section 2 presents the model. Section 3 characterizes extractive social contracts, both in the myopic benchmark and when players are forward looking. Section 4 introduces renegotiation proofness and presents the main result. Section 5 describes the two players' preferred contracts. Section 6 discusses historical applications, comparative statics, and the model's main assumptions. Section 7 concludes. Proofs and additional details are relegated to the Appendix.

Literature Review and Contribution This paper contributes to theory of the state. A recent survey of this literature ([Heldring \[2025\]](#)) organizes the main existing contributions along a "fundamental dichotomy" between two distinct streams of research.

According to the *cooperative* tradition, the state emerged as a solution to collective action problems. Individuals voluntarily surrendered part of their freedom to a sovereign entity in exchange for the provision of public goods, such as the enforcement of property rights (e.g., [Hobbes \[1651\]](#); [Locke \[1689\]](#); [Rousseau \[1762\]](#); [Nozick \[1974\]](#)). For recent

formal contributions on the social contract as a self-enforcing agreement, see [Myerson \[2008\]](#), [Fearon \[2011\]](#), and [Tyson \[2020, 2023\]](#), among others.

By contrast, the *coercive* theories of the state emphasize the role of force. In this view, the state is an "instrument of expropriation" ([Marx and Engels \[1848\]](#)), controlled by the elite to extract surplus from the governed. In [Olson \[1993\]](#)'s famous framework, states formed as warrior elites found it profitable to settle, claim their coercive power over a territory, and extract rents from its inhabitants on a regular basis. A key implication is that because these "stationary bandits" expect to remain in power, they have an incentive to limit extraction in order to preserve the productive capacity of the economy. This observation closely resonates with the main ideas behind my model. I complement it by arguing that rulers reduce rent extraction *only* after the economy has reached a certain level of productivity.

Recent empirical research has provided evidence consistent with both traditions (see [Sánchez de la Sierra \[2020\]](#), [Mayshar et al. \[2022\]](#), and [Allen et al. \[2023\]](#)). My model is agnostic on the origin of the state, which I interpret as the existing relationship between the citizen and the ruler. I do not model how the ruler *takes* power (i.e., what made it possible for him to coercively extract a fraction of output). Instead, I take this as a starting point, and study how the ruler *exercises* power. In this sense, I do not focus on how states form, but rather on how they evolve over time.

For this reason, the closest paper to mine is [Acemoglu and Robinson \[2023\]](#) (see also the "companion" book, i.e., [Acemoglu and Robinson \[2019\]](#)). They model the dynamic relationship between citizens and rulers as a dynamic contest, and characterize the long run equilibrium type of state in terms of the balance of power between the two players. By contrast, my model emphasizes the on path transition between two political regimes, thereby focusing more on the transitional dynamics than on the long run steady state.

This paper is also related to the literature on democratization and growth (e.g., [Acemoglu and Robinson \[2000\]](#); [Lizzeri and Persico \[2004\]](#); [Cervellati et al. \[2025\]](#)), which studies how economic development shapes the incentives of ruling elites to give up political power and "extend the franchise." In this paper, I do not explicitly model political competition. I focus instead on the relationship between growth and rent extraction.

2 Model

A citizen (she) and a ruler (he) repeatedly interact over time. The citizen owns a stock of capital of value $k > 0$, which in every period produces some output according to the following production function:

$$y_t = A_t k,$$

where $A_t > 0$ is the level of productivity at the beginning of period t , and y_t is the level of output.

Time is discrete and runs over an infinite horizon. In every period $t = 1, 2, \dots$, the citizen and the ruler play the following stage game.

The Stage Game The ruler moves first. He announces a tax rate, $\tau_t \in [0, 1]$. The citizen observes it, and chooses one of three actions: $a_t \in \{Q, 0, 1\}$. If the citizen chooses $a_t = Q$, the game ends. The citizen's payoff is equal to the outside option $\bar{u} > 0$, and the ruler's payoff is zero. If instead $a_t \neq Q$, the game continues. In this case, I say that the citizen exerts effort if $a_t = 1$, and does not exert effort if $a_t = 0$.

The citizen's choice of effort determines the growth rate of the economy. In particular,

$$\frac{A_{t+1}}{A_t} = g_{a_t} , \tag{1}$$

where $g_1 > g_0 > 1$, and the initial condition, $A_1 > 0$, is given.²

At the end of each period, output is produced, players' per-period payoffs are realized, and the state variable updates according to (1).

Payoffs Suppose the history of the game reaches period t . The citizen's per-period level of consumption (net of the cost of effort, $c > 0$) is:

$$x_{C,t} = (1 - \tau_t) y_t - c \mathbf{1}\{a_t = 1\} .$$

²New ideas or techniques either arrive exogenously or are discovered locally. For example, tomatoes and potatoes arrived in Europe from the New World in the 16th century. On the contrary, European farmers gradually discovered and improved agricultural techniques, such as crop rotation. One can then interpret g_0 as the exogenous rate at which new ideas arrive from abroad, and g_1 as the faster growth rate achievable when the citizen actively seeks and implements new ideas or techniques.

The ruler's per-period level of consumption is:

$$x_{R,t} = \tau_t y_t .$$

Both players maximize the discounted sum of their life-time consumption flows. The common discount factor is $\delta \in (0, 1)$. I assume $\delta g_1 < 1$, so that discounted payoffs are finite.

Notation and Equilibrium Concept Let $H_t := \{\tau_r, a_r\}_{r=1, \dots, t-1}$ denote the history of the game up to the beginning of period t , and let $A_t(H_t)$ denote the level of productivity (i.e., the state) implied by history H through the sequence $\{a_t\}$. For a strategy profile σ and at a given history H_t , let $W(\sigma|H_t)$ and $V(\sigma|H_t)$ denote the citizen's and the ruler's expected life-time payoffs, respectively. Throughout the remainder of the paper, I often omit time subscripts whenever they are not strictly necessary.

Social contracts (i.e., agreements between rulers and citizens) must be self-enforcing, as players naturally lack commitment power. For this reason, I posit the following definition:

Definition 1 (Social Contracts). *A social contract is a profile of strategies which constitutes a Subgame Perfect Equilibrium of the game. A social contract, σ , is non-extractive at a given history H if the citizen's continuation value is strictly higher than the outside option:*

$$W(\sigma|H) > \bar{u} .$$

Otherwise, it is extractive.

I now characterize the set of self-enforcing social contracts.

3 Preliminary Results: Extractive Contracts

3.1 The Myopic Benchmark

To begin with, suppose players are myopic ($\delta = 0$), or, equivalently, that the game is played only once. Players only care about their current period payoff. The citizen's effort raises output only in the future. Therefore, a myopic citizen never exerts effort. Anticipating this, the ruler's optimal strategy is to maximize rent extraction, subject to the citizen's participation constraint.

In equilibrium, the citizen's participation constraint binds. The ruler can never afford to let the citizen quit the game, as he would otherwise obtain a payoff of zero. This endogenously constrains the ruler's extractive power. The citizen does not quit the game if $(1 - \tau)kA \geq \bar{u}$. Therefore, the ruler's optimal tax rate is:

$$\tau = 1 - \frac{\bar{u}}{kA} := \tau^m(A) . \quad (2)$$

The next statement follows.

Proposition 1 (The Myopic Social Contract is Extractive). *Suppose $\delta = 0$. Every social contract is extractive at every on path history.*

In the following section, I show that a (qualitatively) analogous social contract can emerge also when players are forward looking.

3.2 The Extractive Social Contract

Definition 1 requires that, under any permanently extractive social contract, the citizen's continuation value is equal to the outside option at every history.

Suppose such a contract exists. Since the citizen's flow payoff, x_C , is then constant over time,

$$\bar{u} = x_C + \delta \bar{u} \Rightarrow x_C = (1 - \delta) \bar{u} := \phi .$$

To leave the citizen a flow payoff of ϕ (and, thereby, a constant continuation value of $\bar{u}$),

the tax rate announced by the ruler satisfies:³

$$(1 - \tau) Ak = \phi \Rightarrow \tau = 1 - \frac{\phi}{kA} := \tau^e(A) . \quad (3)$$

I now show that such a tax rate is implementable in equilibrium. Let σ^e denote the extractive social contract, defined as follows. At every state A , the ruler plays $\tau^e(A)$. The citizen stays and does not exert effort ($a = 0$) if and only if $\tau \leq \tau^e(A)$; otherwise, she quits ($a = Q$).

Proposition 2 (The ‘Extractive’ Lower Bounds). *The profile of strategies σ^e is a social contract. Under this contract, the ruler’s value function is:*

$$V^e(A) := \frac{kA}{1 - \delta g_0} - \bar{u} > 0 ,$$

and the citizen’s value function is $\bar{u}$. Then, σ^e is extractive at every on path history. Moreover, for any social contract σ and at every (on or off path) history H :

$$V(\sigma|H) \geq V^e(A(H)) \quad \text{and} \quad W(\sigma|H) \geq \bar{u} . \quad (4)$$

Proposition 2 provides an illustration of the prominent argument according to which non-extractive institutions are a "fundamental cause" of long run growth ([Acemoglu et al. \[2001, 2005\]](#)).

The citizen bears the full cost of effort but only internalizes a fraction of the additional output it generates. Therefore, her incentives to exert effort — which increases the growth rate of the economy — depend on the rent extraction rate implemented by the ruler in the future. It follows that the citizen exerts effort if and only if she *expects* rent extraction in the future to be low enough. Under an extractive social contract, these expectations are never fulfilled. Knowing this, the citizen never exerts effort, and growth never takes off.

If $\delta = 0$, the extractive tax rate, (3), and the equilibrium tax rate of the one-shot

³I assume $kA_1 \geq \phi$, so that (3) is always lower or equal than one.

game, (2), coincide. If $\delta > 0$, instead, the extractive tax rate is harsher for the citizen: $\tau^e(A) > \tau^m(A)$. The reason is that the binding constraint for the ruler is the citizen's *life-time* participation constraint. A myopic citizen must be paid her entire outside option in every period. A patient citizen, instead, accepts a strictly smaller flow payoff in every period, $\phi = (1 - \delta)\bar{u} < \bar{u}$, because she anticipates "the value of the future."

Proposition 2 also implies that:

Remark 1. *Under any social contract, the citizen never quits: $a_t \neq Q$ at every on path history.*

As discussed in the Introduction, the existence of a relational contract between the citizen and the ruler is a necessary (but not sufficient) condition for sustained growth. In other words, the economy can grow at the fast rate, g_1 , only if a social contract is in place; and if a social contract is in place, the citizen never quits, so that sustained effort and growth is possible (even if not automatic, as proved by Proposition 2).

Finally, note that:

Remark 2. *σ^e is the (essentially) unique social contract which is extractive at every on path history.*

The argument supporting this remark is simple. If a social contract is extractive at every on path history, the citizen's continuation value is equal to $\bar{u}$ along the whole path. This univocally pins down the equilibrium tax rate, $\tau^e(A_t)$ — as shown above. The contract is therefore "essentially" unique because while on path play is univocally characterized, different contracts may support this equilibrium play through different off path punishments.

Figure 2 plots the paths of the equilibrium tax rate and of the two players' value functions under the extractive social contract.

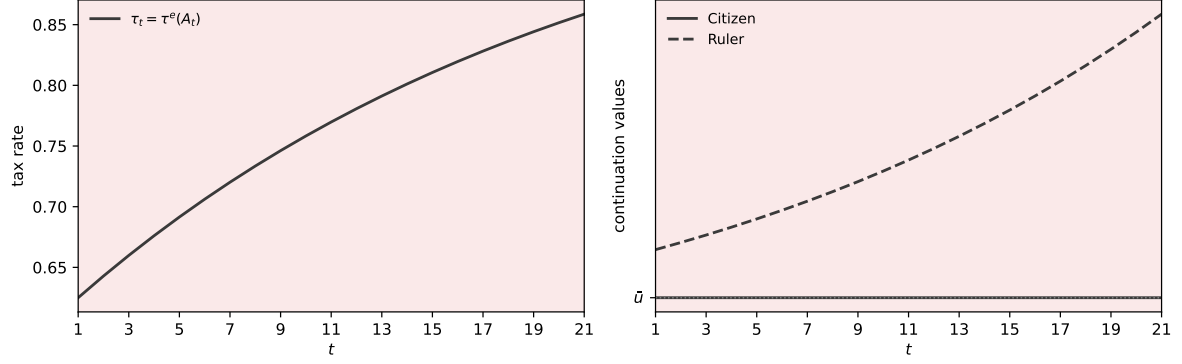


Figure 2: **The extractive social contract:** Equilibrium tax rate (left), and players' continuation values (right). The background indicates *extractiveness* of the contract: green where the contract is non-extractive, and red where it is extractive.

Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c = 0.5$, $g_0 = 1.05$, $g_1 = 1.15$, $A_1 = 0.8$.

To sum up, while extraction is the unique possible outcome if players are short-lived, it can also arise when they are not. Patience, therefore, does not reduce extraction. However, it also opens the possibility of *promises*.

In this sense:

Proposition 3 (Any Markovian Social Contract is Extractive). *In any social contract which is non-extractive at a given history H , players' strategies are a function of the history of the game.*

4 Renegotiation Robust Contracts

Proposition 3 implies that non-extractive social contracts need to be sustained by promises.

Along any of these contracts, the ruler *buys* the citizen's effort in period t with a promise about lower taxes in period $t + 1$. The resulting value for society is the extra surplus (net of effort costs) generated when the citizen exerts effort in every period rather than never.

To compute this value, let

$$\Pi_0(A) := \sum_{t \geq 0} \delta^t k A g_0^t = \frac{kA}{1 - \delta g_0}, \quad \Pi_1(A) := \sum_{t \geq 0} \delta^t (k A g_1^t - c) = \frac{kA}{1 - \delta g_1} - \frac{c}{1 - \delta} \quad (5)$$

denote society's surplus, starting from state A , when the citizen never exerts effort and when she exerts effort in every period, respectively. The value of the chain of promises is the difference:

$$\Delta(A) := \Pi_1(A) - \Pi_0(A) = \frac{\delta(g_1 - g_0)kA}{(1 - \delta g_1)(1 - \delta g_0)} - \frac{c}{1 - \delta}, \quad (6)$$

which is linear increasing in A .

These *purchases* have a price. Consider a contract, σ , which prescribes the citizen's effort at some history H . The tax rate τ has already been announced, so the citizen's current consumption flow is the same whether or not she exerts effort. Withholding effort therefore saves her the cost c today, and, by Proposition 2, still secures a continuation value of at least $\bar{u}$ from period $t + 1$ onwards. Effort is thus incentive compatible only if

$$-c + \delta W(\sigma|H_{t+1}) \geq \delta \bar{u}, \quad (7)$$

that is, only if the contract promises the citizen a continuation value of at least

$$\omega := \bar{u} + \frac{c}{\delta} > \bar{u} \quad (8)$$

from the next period onwards.

Social contracts must also prescribe punishments for those who do not keep their promises. By subgame perfection, punishments must be equilibria of the continuation (off path) game themselves. However, not all self-enforcing punishments are equally deterrent. For instance, when entering a punishment phase in period t , after a deviation has occurred in period $t - 1$, players may jointly abandon the previously agreed punishment in favor of another self-enforcing agreement that both prefer. The literature refers to equilibria sustained by such possibly implausible punishments as not *renegotiation proof* (see Farrell and Maskin [1989]; Levin [2003] or Acemoglu et al. [2008] for a political economy application).

The goal of this section is to characterize the whole set of renegotiation proof social contracts.

Formally,

Definition 2 (Renegotiation Proof Social Contracts). *A social contract σ is renegotiation proof (RP) if, at the beginning of every period, after any (on or off path) history H , there is no social contract of the continuation game, σ' , such that:*

$$W(\sigma'|H) > W(\sigma|H) \quad \text{and} \quad V(\sigma'|H) > V(\sigma|H) .$$

Having stated a formal definition for renegotiation proofness, I am now ready to study its political economy consequences.

For any state A , let $\Sigma(A)$ denote the set of RP social contracts of the game with initial state A . My main result is:

Proposition 4 (The Dual Transition). *There exists a calendar period $T^* \geq 2$ such that, for every $\sigma \in \Sigma(A_1)$, on the equilibrium path, the following properties hold.*

- (i) *The growth rate of the economy is g_0 before T^* , and g_1 from T^* onwards.*
- (ii) *The citizen's continuation value, $W(\sigma|H_t)$, is such that:*

$$W(\sigma|H_t) \in \begin{cases} [\bar{u} , \bar{u} + \delta^{T^*-t} \Delta(A_{T^*})] & \text{if } t \leq T^* , \\ [\omega , \bar{u} + \Delta(A_t)] & \text{if } t > T^* , \end{cases} \quad (9)$$

and the equilibrium tax rate is pinned down by the citizen's Bellman Equation:

$$(1 - \tau_t) k A_t = W(\sigma|H_t) - \delta W(\sigma|H_{t+1}) + c \mathbf{1}\{t \geq T^*\} . \quad (10)$$

- (iii) *The contract is non-extractive at every on path history with $t > T^*$. As a consequence, taxation cannot remain extractive: $\tau_t < \tau^e(A_t)$ at some $t > T^*$.*

Moreover, the set of RP social contracts is non-empty: $\Sigma(A_1) \neq \emptyset$.

The proof is presented in Appendix A. I now briefly sketch its logic here. It proceeds in two main steps. The first step characterizes the largest total surplus that equilibrium play can generate from any state A . I denote this quantity by $\bar{\Pi}(A)$ (see Definition 3, in the Appendix). Note that the maximum level of surplus attainable in equilibrium, $\bar{\Pi}(A)$, need not coincide with the value function of a social planner's problem. In this sense, it is a *second best* measure of efficiency.

Lemma 1 first shows that, in any surplus-maximizing plan, once effort starts, it never stops. Building on this, I show that $\bar{\Pi}(A)$ is attained by a plan characterized by a simple 'threshold rule', according to which the citizen does not exert effort until the level of productivity reaches a certain threshold, A^* , and exerts effort forever thereafter (Lemma 2). The calendar period T^* is the time at which an economy growing at rate g_0 reaches A^* .

The second step connects renegotiation proofness with second best efficiency. The key observation is that the tax rate implemented by the ruler does not affect total surplus, but only how surplus is split between the two players. Hence, two contracts that prescribe the same path of effort generate the same surplus, and differ only in its division. Building on this, Lemma 3 shows that every division of $\bar{\Pi}(A)$ which respects the two lower bounds characterized in Proposition 2 is attainable by a social contract. It follows that any RP social contract must be such that players' value functions sum to $\bar{\Pi}(A)$ at every on or off path history (Lemma 4). Finally, Lemma 5 delivers existence: $\Sigma(A_1) \neq \emptyset$.

Proposition 4 describes the equilibrium path of institutional change. At the beginning of every on path history, the economy stagnates. After a certain calendar period is reached, economic growth accelerates permanently. The economic boom is followed by a permanent tax concession the ruler grants to the citizen.

In this regard, the size of the tax concession deserves a remark:

Remark 3. *At every $t > T^*$, the present value of the citizen's consumption plan along any RP social contract exceeds $\bar{u} + \frac{c}{1-\delta}$ (i.e., the outside option plus the full compensation for the cost of repeated effort) by at least $K := \frac{c}{\delta}$.*

Why can't the ruler simply compensate the cost of effort? The answer comes from the dynamic nature of the relationship. Sustained growth comes from the citizen's repeated effort. This can be seen as a sequence of one-period "loans" from the citizen to the ruler. In fact, the citizen pays the cost of effort in period t and is rewarded only in period $t + 1$.

For the loan to be optimal for the lender, the repayment must cover the principal, c , plus the interest. The net interest per loan is $\frac{c}{\delta} - c = \frac{c(1-\delta)}{\delta}$. The *rent* K capitalizes this "perpetual interest," as the present value of infinitely many of such payments is exactly $\frac{c}{\delta}$. To summarize, because the growth process is dynamic, the citizen is a strict beneficiary of growth rather than merely reimbursed for producing it.

The ruler's trade-off is also intertemporal. If he fails to repay the debt (i.e., if he raises the tax rate in period t when the citizen had exerted effort in period $t - 1$), he would increase his current period payoff. However, the citizen might punish this behavior (see the proof of Lemma 3, in Appendix B, for an explicit construction of the punishments). Therefore, the ruler's restraint from eating more pie today is the price he has to pay in order to have a larger pie tomorrow.⁴

As already discussed, on the path of any RP social contract, equilibrium social welfare is maximized, meaning that no other equilibrium can deliver a higher surplus (Lemma 4). In particular, society attains the second best level of welfare when the citizen exerts effort after $T^* - 1$ periods from the beginning of the game.

By contrast, a first best plan could prescribe the citizen to start exerting effort before the transition (i.e., starting at some $t' < T^*$), even if this may not be jointly incentive compatible for the two players. Incentive compatibility can therefore postpone the transition. The first best allocation is however itself a threshold rule, prescribing no effort until productivity is high enough, and effort in every period thereafter. Even a social planner unconstrained by incentives would let the economy stagnate first.

Formally, A^* is the larger of two thresholds. The first, $\hat{A}$, is an *efficiency* threshold. Above this level of productivity, starting a sequence of sustained effort is socially desirable. The second, $\tilde{A}$, is an *incentive compatibility* threshold. Above it, the surplus

⁴I return to players' incentive compatibility constraints when focusing on their two best preferred contracts, in Section 5.

created by effort is large enough to cover both the citizen's premium, $\frac{c}{\delta}$, and the ruler's lower bound, $V^e(A)$. Before the state reaches A^* , then, effort fails either because no equilibrium can reward it, or because rewarding it is inefficient.

The key intuition behind this result is the complementarity between existing and new technologies, or ideas. Formally, the citizen's effort at state A raises next period's productivity from Ag_0 to Ag_1 . The *additional* output it generates in the following period is then equal to $kA(g_1 - g_0)$. The benefit of effort is thus proportional to the level of productivity already attained, while its cost, c , is not. In this sense, I say that the citizen's effort exhibits a "standing on shoulders" property. The return to discovering or implementing a new idea is proportional to the stock of ideas already in use (see Section 6).

At higher levels of productivity, raising rent extraction above the level at which the citizen is willing to exert effort implies a larger output loss for society. For this reason, the ruler also prefers a permanently lower share of a growing pie, rather than the whole of a stagnant one.

Statement (ii) also shows that, while total surplus is necessarily maximized on the path of any RP social contract, its *distribution* is not univocally pinned down. In particular, Equation (9) characterizes a "band" of equilibrium payoffs. Every point of the band is attainable by some RP social contract (Lemma 5). Given the whole path of the citizen's value function, Equation (10) pins down the corresponding path of announced tax rates. Since the two players' value functions always sum up to the same value, the dynamics of the citizen's value function univocally characterizes the entire contract (on path).

In equilibrium, the lower bound on the citizen's life-time payoff at a given history depends on the distance in time to the transition, T^* . Before the institutional transition, the floor is simply equal to the outside option. After the transition, the citizen's floor increases (see Remark 3). The ceilings are instead obtained as lower bounds on the ruler's life-time payoff. The citizen can never be promised more than the extra surplus that sustained growth generates.

The "width" of the band thus changes over time. In particular, very far from the transition, *every* renegotiation robust social contract is extractive. Figure 3 plots the band. Its two boundaries correspond to the players' preferred contracts (σ_C^*, σ_R^*) . Section 5 characterizes them in detail. Every other RP social contract lies in between.

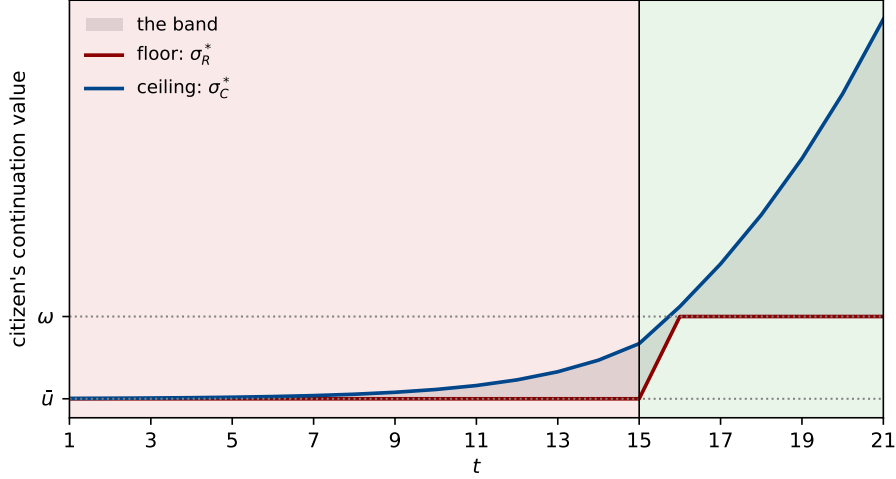


Figure 3: **The citizen's value function under any RP social contract.** The background indicates *extractiveness*: red where extraction is still possible, and green where it is ruled out for every RP social contract.

Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c = 0.5$, $g_0 = 1.05$, $g_1 = 1.15$, $A_1 = 0.8$, which implies: $\bar{A} > \bar{A}$, $A^* \approx 1.53$, and $T^* = 15$ (vertical line).

Because any point on the Pareto frontier (i.e., the band) is attainable by a social contract, social welfare is maximized also off path.

Two further remarks follow. First:

Remark 4. *The extractive social contract is not renegotiation proof, i.e., $\sigma^e \notin \Sigma(A_1)$.*

Under the extractive social contract, the growth rate of the economy is minimized. By contrast, Lemma 2 shows that sustained growth is achievable after T^* . Therefore, whenever the two players find themselves at a history whereby effort would be Pareto efficient, no RP social contract can prescribe extraction. Players would renegotiate that.

Therefore,

Remark 5. *Consider any social contract that punishes some deviation by permanent reversion to σ^e . Then the contract is not renegotiation proof.*

In other words, a non-extractive, renegotiation robust social contract cannot be sustained by the threat of a permanent revert to the extractive equilibrium (i.e., by grim trigger strategies).

To further characterize punishments which satisfy RP, I focus on the two players' best contracts.

5 The Two Players' Best Contracts

Proposition 4 characterizes what *every* RP social contract must deliver. This section describes two examples and focuses on the two players' ex ante optimal contracts. The two players' payoff along the path of these two contracts correspond to the two boundaries of the band shown in Figure 3.

Both contracts are constructed formally in the proof of Lemma 3, in Appendix B. In what follows, I briefly summarize their off and on path prescriptions.

Off path, the two contracts serve as each other's punishment. If the ruler deviates in period t , play follows σ_C^* from period $t + 1$ onwards. If the citizen deviates, play follows σ_R^* . This is a consequence of renegotiation proofness. By Remark 5, punishments cannot shrink the pie. The only available threat to sustain "cooperation" along a RP contract is a different split of the same pie.⁵

On path, both contracts prescribe the same actions for the citizen. She exerts effort if and only if $A \geq A^*$, and stays in the game if and only if $\tau \leq \tau^e(A)$. The two contracts differ only in the tax rate implemented by the ruler, and therefore in how the surplus generated by growth is split between the two players.

⁵The contracts also prescribe that the citizen "punish" the ruler and withdraws effort in the period of a ruler's deviation; see Remark 9.

The ruler-preferred contract Under σ_R^* , the citizen's value function is equal to the lower bound. The ruler receives the remainder of the surplus. Therefore,

$$W(\sigma_R^*|H_t) = \begin{cases} \bar{u} & \text{for } t \leq T^* , \\ \omega & \text{for } t > T^* , \end{cases} \quad V(\sigma_R^*|H_t) = \bar{\Pi}(A_t) - W(\sigma_R^*|H_t) .$$

The implied tax rates are:

$$\tau_t = \tau^e(A_t) \quad \text{for } t \leq T^* , \quad \tau_t = 1 - \frac{\phi + c/\delta}{kA_t} \quad \text{for } t > T^* .$$

Therefore, the ruler's preferred contract is extractive at every period $t \in [1, T^*]$, and non-extractive forever thereafter.

Figure 4 plots the equilibrium tax rate, and the two players' value functions along this contract.

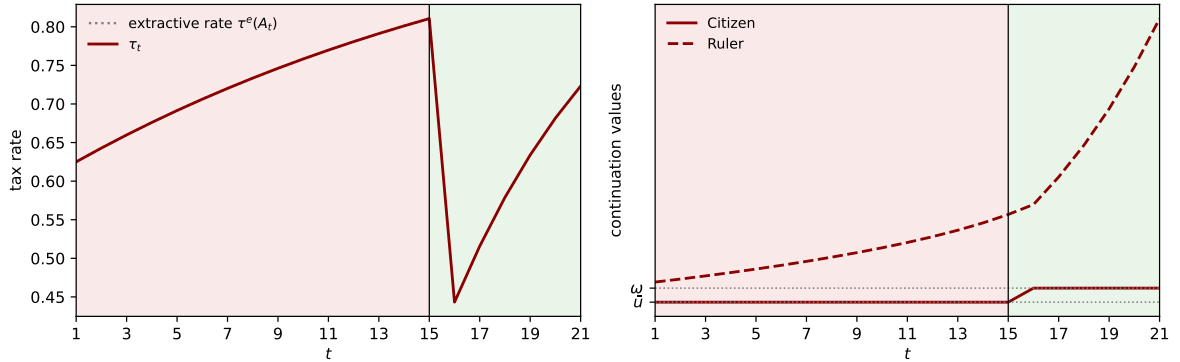


Figure 4: **The ruler-preferred contract:** Equilibrium tax rate (left), and players' continuation values (right). The background indicates *extractiveness* of the contract: green where the contract is non-extractive, and red where it is extractive.

Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c = 0.5$, $g_0 = 1.05$, $g_1 = 1.15$, $A_1 = 0.8$, which implies: $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$ (vertical line).

The citizen-preferred contract Under σ_C^* , the citizen's value function is equal to the upper bound. The ruler receives the remainder of the surplus:

$$W(\sigma_C^*|H_t) = \begin{cases} \bar{u} + \delta^{T^*-t} \Delta(A_{T^*}) & \text{for } t < T^* , \\ \bar{u} + \Delta(A_t) & \text{for } t \geq T^* , \end{cases} \quad V(\sigma_C^*|H_t) = V^e(A_t) .$$

The corresponding tax rates are:

$$\tau_t = \tau^e(A_t) \quad \text{for } t < T^*, \quad \tau_t = \tau^e(A_t) - \frac{\delta(g_1 - g_0)}{1 - \delta g_0} \quad \text{for } t \geq T^* .$$

In this case, the citizen's rent (the difference between her value function and the outside option) is strictly positive at every period. Hence, σ_C^* is non-extractive at *every* on path history.

Figure 5 summarizes this social contract.

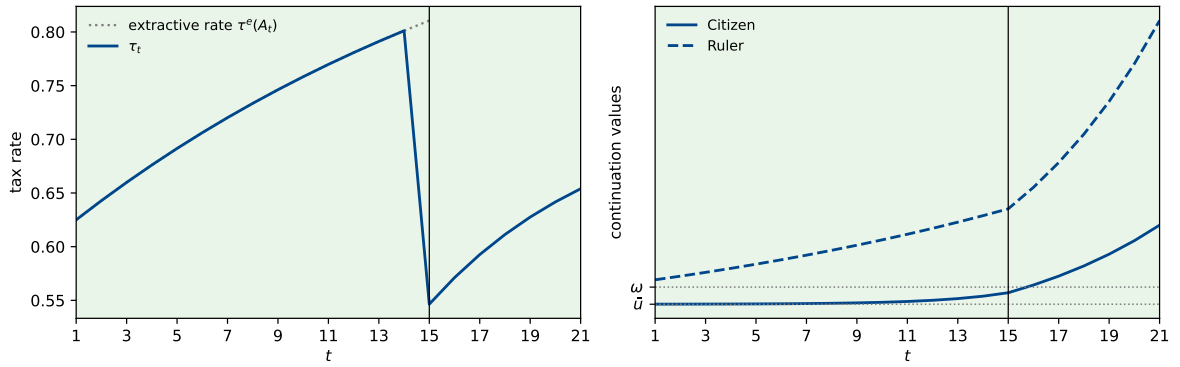


Figure 5: **The citizen-preferred contract:** Equilibrium tax rate (left), and players' continuation values (right). The background indicates *extractiveness* of the contract: green where the contract is non-extractive, and red where it is extractive.

Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c = 0.5$, $g_0 = 1.05$, $g_1 = 1.15$, $A_1 = 0.8$, which implies: $\tilde{A} > \hat{A}$, $A^* \approx 1.53$, and $T^* = 15$ (vertical line).

5.1 Comparing the Two Contracts: On Path

Before the transition (for every $t < T^*$), the two contracts characterized in this section are observationally equivalent. They prescribe the same equilibrium tax rate, which also coincides with the tax rate prescribed by the extractive social contract derived in Proposition 2.

However, the citizen's preferred contract is always non-extractive, meaning that the citizen finds it always *strictly* optimal to stay in the game. This happens because, even before the transition, the citizen anticipates the values of the transition itself. By contrast, the ruler's preferred contract is non-extractive only after the transition. In this case, the ruler concedes the lowest tax cut that makes effort incentive compatible for the

citizen. In present value terms, at any $t < T^*$, the transition is therefore worth nothing to the citizen, and her value function is equal to the outside option.

At the moment of the transition from stagnation to growth, both contracts strictly benefit the citizen with respect to what would happen in an extractive equilibrium. This is also optimal for the ruler, who anticipates that, by reducing his share of the pie today, he will allow the pie to grow faster, and thus be able to eat more pie in the future. In this sense, the ruler's patience is a key determinant of the transition.

To see this, consider the ruler's different consumption plans. In any period, along any contract, the ruler consumes all the output that he does not leave to the citizen, i.e.,

$$x_{R,t} = kA_t - x_{C,t} - c \mathbf{1}\{a_t = 1\} .$$

Along σ^e , the citizen never exerts effort, so the level of productivity in period t is $A_1 g_0^{t-1}$. The ruler's flow of consumption is $kA_1 g_0^{t-1} - \phi$. Along *any* RP social contract, the ruler's consumption exceeds this benchmark by:

$$\underbrace{k(A_t - A_1 g_0^{t-1})}_{\text{extra output}} - \underbrace{[x_{C,t} - \phi + c \mathbf{1}\{a_t = 1\}]}_{\text{"tax concession"}} .$$

Focusing on σ_R^* and σ_C^* , we can see that before T^* , both terms are equal to zero. After T^* , both terms are positive.

Under σ_R^* , the tax concession is minimized and it is equal to the constant K , while the "extra output" term grows unboundedly as the state increases. Therefore, there exists a finite calendar period after which the latter term dominates the former. Under σ_C^* , both terms are instead linear in the state. The tax concession term increases linearly with constant slope $\frac{\delta k(g_1 - g_0)}{1 - \delta g_0}$ (see also below). The extra output term is also linear, but its slope increases over time.⁶

Therefore:

⁶To see this, write

$$A_t = A_{T^*} g_1^{t-T^*} , \quad A_1 g_0^{t-1} = A_{T^*} g_0^{t-T^*} ,$$

so that:

$$k(A_t - A_1 g_0^{t-1}) = kA_{T^*} (g_1^{t-T^*} - g_0^{t-T^*}) = kA_t \left[1 - (g_0/g_1)^{t-T^*} \right] .$$

Remark 6. Consider the two players' preferred contracts, i.e., σ_R^* and σ_C^* . Under both contracts, the ruler's consumption flow is lower than under the extractive contract for finitely many periods after T^* , and it is higher forever thereafter.

Figure 6 illustrates this remark.

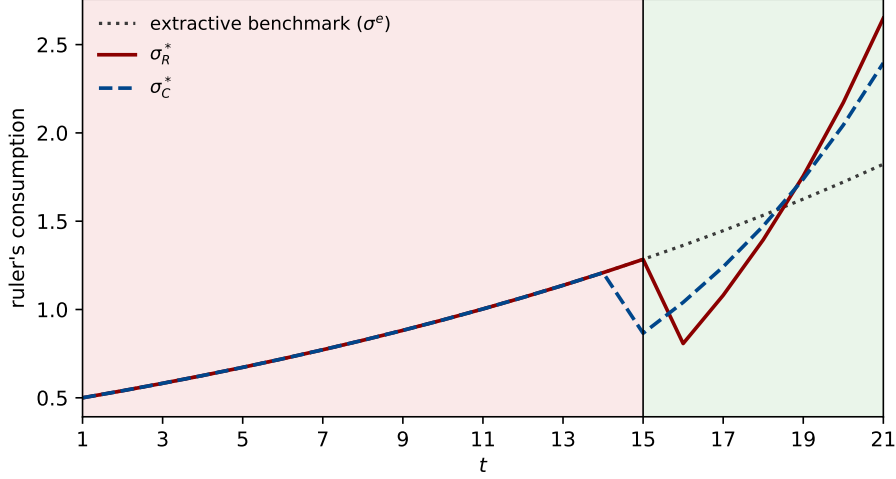


Figure 6: **The ruler's consumption flow along the two players' preferred contracts.** The background indicates *extractiveness*: green where extraction is ruled out for every RP social contract, and red where it is still possible.

Note: Parameters: $\delta = 0.7$, $k = 1$, $\bar{u} = 1$, $c = 0.5$, $g_0 = 1.05$, $g_1 = 1.15$, $A_1 = 0.8$, which implies: $\bar{A} > \bar{A}$, $A^* \approx 1.53$, and $T^* = 15$ (vertical line).

Another difference between the two contracts is that they deliver the tax concession to the citizen at different dates, and in different ways. Under σ_R^* , the ruler gives up a fixed *amount* of the pie in every period after the transition, and thus the citizen's rent is constant. As already emphasized, this amount is the lowest possible which makes it incentive compatible for the citizen to exert effort. By contrast, under σ_C^* , the ruler gives up a fixed *share* of output, $\frac{\delta(g_1 - g_0)}{1 - \delta g_0}$, and the citizen's rent, $\Delta(A)$, grows over time.

As a consequence, the comparison sharpens in the long run:

Remark 7. As $A \rightarrow \infty$, the tax rate prescribed by σ_R^* converges to the extractive rate $\tau^e(A)$. The tax rate prescribed by σ_C^* converges instead to

$$1 - \frac{\delta(g_1 - g_0)}{1 - \delta g_0} < 1.$$

Its slope, $k \left[1 - (g_0/g_1)^{t-T^*} \right]$, is zero at T^* and increases towards k . It therefore eventually exceeds the slope of the tax concession term, which is lower than k because $\delta g_1 < 1$.

Finally, notice that, by Statement (ii) of Proposition 4:⁷

Remark 8. *The social contract σ_R^* (respectively, σ_C^*) is the (essentially) unique RP social contract under which the citizen's (respectively, the ruler's) value function is equal to its lower bound at every on path history.*

Notice that Remark 8 does not imply that no other ex ante optimal contracts exist for the two players.

For example, consider the ruler's ex ante preferred contract, σ_R^* . This definition only requires $W(\sigma_R^*|H_1) = \bar{u}$, which pins down the period $t = 1$ split of the surplus. Alternative contracts could prescribe the ruler to concede slightly more to the citizen today in exchange for slightly less tomorrow, leaving $W(\sigma_R^*|H_1)$ (and thus his own payoff) unchanged. Any such contract implements a different path of tax rates. Uniqueness therefore follows from a stronger requirement, i.e., that the citizen's continuation value is equal to its lower bound at *every* history. The same holds for σ_C^* .

5.2 Comparing the Two Contracts: Off Path

I now discuss why neither player finds it appealing to deviate from the contracts described in this section. The logic of the punishments follows directly from renegotiation proofness.

Consider the ruler first. His most tempting deviation is to increase the tax rate once the citizen has already exerted effort. How is that not optimal?

The punishment which deters such behavior is the loss of all future bargaining power. The citizen withdraws effort for one period, and play follows σ_C^* thereafter, which pushes the ruler down to his lower bound, $V^e(A)$. By deviating, the ruler collects at most $kA - \phi$ in tax revenues in the current period. He can never secure more than the whole output, net of the minimal payment that keeps the citizen in the game (see Section 3.2). Along the deviation, his continuation value is equal to $V^e(Ag_0)$, because the citizen withdraws effort in the current period. Summing the two terms, $kA - \phi + \delta V^e(Ag_0) = V^e(A)$, so that the deviation can never be strictly profitable.

⁷See Remark 2 for a discussion on *essential* uniqueness of the extractive contract.

This last statement follows from the ruler's Bellman equation along the extractive contract. His lower bound, $V^e(A)$ is, by construction, the value of extracting all available resources in every period, when the citizen never exerts effort. By deviating, the ruler replicates exactly this contract. He extracts everything for one period, the economy stagnates, and the punishment holds him to his extractive value forever after. In other words, the deviation turns him into the extractive ruler of Section 3.2.⁸

This argument relies on how the citizen responds to the ruler's deviation:

Remark 9. *Consider the off path prescriptions of the two players' preferred contracts, i.e., σ_R^* and σ_C^* . If the citizen keeps exerting effort following a ruler's deviation, the deviation would be worth $kA - \phi + \delta V^e(Ag_1) > V^e(A)$, and it would be strictly profitable.*

Consider now the citizen's choice to cooperate. The most tempting deviation available for her is to withdraw effort once the ruler has already announced a low tax rate. By not exerting effort, the citizen increases her current period payoff. The punishment prescribes that the contract σ_R^* is in place starting the next period. This contract is restarted at a phase in which no effort has been exerted, which pushes the citizen's continuation value down to her lower bound, $\bar{u}$. By deviating, the citizen thus loses the premium $\omega - \bar{u} = \frac{c}{\delta}$ that the contract promises after every period of effort. Since this loss exceeds the saved cost c (as $\delta > 0$), withdrawing effort is not profitable.

Before concluding, I now briefly return to the general properties and results of the model, and discuss historical applications.

6 Discussion: The *Dual* Transition

6.1 Historical Examples

The key prediction of the model is that (on the path of any RP equilibrium) society necessarily experiences a *dual transition*. Up to a certain calendar period, the economy stagnates and extractive social contracts are implementable. From that period onwards, growth accelerates permanently, and taxation cannot remain extractive.

⁸This is shown formally in Step 2 of the proof of Lemma 3 (Appendix B).

The idea of a dual transition closely resembles the joint political and economic transformation experienced by Europe between 1789 and 1848 — a process which [Hobsbawm \[1962\]](#) famously described as a "dual revolution." More in general, my theoretical results may offer an explanation for the evolution of institutions observed over the very long run. For most of recorded history, the typical society resembled the model's pre-transition phase. Output per person barely grew, and whatever surplus produced beyond subsistence needs was captured by narrow ruling elites through taxes or forced labor. Over the very long run, however, human societies have tended (albeit slowly, unevenly, and often not permanently) towards sustained economic growth and less extractive and more accountable institutions. The model's central message is that these two tendencies are not parallel events, but two halves of a single, dual transition — exactly as argued by [Hobsbawm \[1962\]](#).

Medieval and early modern England offers a well-documented illustration of both halves of the transition. Consider first the institutional one. For much of the medieval period, the king of England could tax and confiscate his subjects' resources at his discretion. In the twelfth century, the crown built one of Europe's first centralized fiscal systems, auditing local sheriffs directly from London through the *Exchequer* ([Strayer \[1970\]](#)).⁹ As early as 1215, the *Magna Carta Libertatum* imposed the first formal limits on the king's authority, requiring nobles' consent for new levies. These requirements, however, were not effectively enforced for centuries. By contrast, the following years were characterized by increased centralization of political power in the hands of the king ([Acemoglu and Robinson \[2012\]](#)). Effective constraints on the crown arrived only four centuries later. The struggle between the lords and the crown turned violent with the English Civil Wars (1642–1651). The Glorious Revolution of 1688 and the Bill of Rights of 1689 ultimately established constitutionally that no tax could be levied without the consent of Parliament. England has been a constitutional monarchy ever since.

The resulting outcome is consistent with the model. Losing the power to set taxes at

⁹It is against this background of centralized and extractive taxation that, over the following centuries, the legend of Robin Hood spread: an outlaw who fights the sheriff and robs the Prince to return tax revenues to ordinary people.

will did not impoverish the crown. By contrast, [North and Weingast \[1989\]](#) documents how, after 1689, tax revenues and state capacity rose sharply. [Acemoglu and Robinson \[2012\]](#) also argues how the emergence of inclusive institutions in England at the end of the 17th century may explain why the industrial revolution started there rather than elsewhere.¹⁰

The economic and institutional paths of other high or middle income countries are similar. In France, the fiscal privileges of the *ancien régime* were abolished in 1789, and constitutional limits on the executive established after the Revolution were followed by sustained industrial growth over the 19th century. In Japan, two centuries of near-stagnation under the Tokugawa shogunate ended with the Meiji Restoration (1868), which dismantled the feudal system, established a constitution (1889), and opened the way to rapid industrialization. In the twentieth century, the sustained growth of South Korea and Taiwan was accompanied, within a generation, by the transition to democracy ([Huntington \[1991\]](#)).

6.2 Comparative Statics

The dual transition, however, did not happen everywhere at the same time. Some societies escaped stagnation and extraction centuries before others, and some are yet to do it today. For example, European polities have adopted systems of checks and balances on the executive power earlier than elsewhere ([Stasavage \[2016\]](#)). Why is that the case? The comparative statics of the model provides insights to understand the sources of this heterogeneity.

One possible explanation emphasizes the role of geography as a cause of Europe's early technological lead ([Diamond \[1997\]](#)). In this view, the availability of domesticable plants and animals, together with the east–west orientation of the Eurasian continent, facilitated the early adoption and diffusion of agricultural technologies across regions with similar climatic conditions.

¹⁰For a recent empirical contribution on the famous debate in economic history on "Why England?," see [Kelly et al. \[2023\]](#), which documents how industrialization occurred in areas of the country characterized by high wages and strong mechanical skills.

In the language of my model, Europe's early exogenous technological advantage may be read as a higher initial condition, A_1 , which implies that the economy reaches the transition threshold, A^* , earlier. Consistent with this argument, [Comin et al. \[2010\]](#) documents that Europe was *already* technologically more advanced than the rest of the world in 1000BC.

A second possible explanation concerns the time-horizon of rulers and length of European reigns. In the model, political stability (which can be read as a higher δ) accelerates the transition. This happens because of two effects. First, because the non-extractive regime generates permanently higher growth, a patient ruler is more willing to sacrifice current rents in exchange for a larger pie tomorrow. Second, the citizen pays the cost of effort one period before she is repaid for it. Therefore, the lowest return to effort which makes it incentive compatible for the citizen, ω , decreases, and the transition becomes cheaper for the ruler.

Consistent with the prediction, there is empirical evidence of a positive statistical association between the length of European reigns and state performance (see [Clague et al. \[1996\]](#) and [Ottinger and Voigtländer \[2025\]](#), Online Appendix, Table A.21, for a more recent contribution).¹¹

6.3 Discussion of the Assumptions

So far, we have seen that, even if the model is necessarily simplistic, its results are consistent with the evolution of economic and political institutions over the very long run. In this paragraph, I briefly comment on what assumptions are needed to obtain these results.

The 'Standing on Shoulders' Property The existence of a dual transition heavily depends on the "standing on shoulders" property of technological change. This property means that the (net) returns to the citizen's effort are proportional to the existing level of productivity.

¹¹The direction of causality is difficult to establish, as better institutions may also contribute to longer and more stable reigns.

To better see the role played by this assumption, suppose, by contrast, that returns to effort do not depend on the level of productivity, i.e., $A_{t+1} = g_0 A_t + B \mathbf{1}\{a_t = 1\}$, for some $B > 0$. The additional output generated by the citizen's effort is a constant. In this case, effort would be socially desirable either in every period or never. For this reason, there would be no transition. The economy would be "locked in" a single regime from the start. In this sense, the complementarity between effort and the existing stock of ideas is the ultimate source of the transition.

The Outside Option The baseline model assumes a constant outside option for the citizen. This assumption captures the idea that if the citizen does not accept the ruler's power, she loses access to capital as property rights are not enforced. However, one may also imagine that the size of the economy determines the outside option, for instance because revolts are more likely to succeed when societies are technologically advanced. One way to accommodate this intuition is by assuming that the outside option is increasing in the state, i.e., $\bar{u}(A) = uA$, for some small enough $u > 0$.

This alternative assumption has two main consequences. First, the extractive tax rate is a constant, $\tau^e = 1 - \frac{u(1-\delta g_0)}{k}$. Second, the minimal continuation value for the citizen in a period of effort, ω , grows with the state. By not exerting effort at state A , the citizen secures a continuation value of $\bar{u}(Ag_0)$, so that $\omega(A) = \bar{u}(Ag_0) + \frac{c}{\delta}$.

The main results, however, are robust to this alternative modeling choice. The social value of effort still increases with A , so the equilibrium still features a threshold rule. The transition, however, arrives earlier. To see why, recall that, to induce effort, the ruler must promise the citizen a profit of $\frac{c}{\delta}$ on top of the value of the outside option $\bar{u}$. The ruler can afford this promise only if the extra surplus generated by growth covers it, i.e., if $\Delta(Ag_1) \geq \frac{c}{\delta}$. This condition defines the threshold $\tilde{A}$. With a state-dependent outside option, the condition becomes $\Delta(Ag_1) \geq \frac{c}{\delta} - uA(g_1 - g_0)$, which is weaker than in the baseline, so that $\tilde{A}$ decreases. The intuition is that effort also increases the citizen's future outside option, which the ruler must always concede. Part of the repayment for effort is thus paid through this channel, and promises are cheaper for the ruler.

7 Conclusions

This paper studies a simple society made of only two agents: a citizen and a ruler. In a simple repeated game, I show that stagnation and extraction cannot last forever. On the path of every Pareto efficient and renegotiation proof social contract, there comes a date at which the citizen starts to exert effort, economic growth takes off, and the ruler concedes a non-extractive tax rate.

The transition is ultimately driven by the accumulation of knowledge. As productivity rises, the citizen's effort becomes more valuable, until the ruler comes to prefer a smaller share of an expanding economy over the entirety of a stagnant one.

The historical implication of this result is that modern rulers are less extractive than ancient ones not because constitutions or other formal institutions constrain them, but because citizens' work has become so productive that discouraging it through extraction is no longer in their own interest. In this view, constitutions are a product of the transition rather than its cause.

The model is simple and deliberately omits many interesting complications. For instance, I do not model political competition between multiple rulers and collective action problems among multiple citizens. These extensions are left for future work. Whether the dual transition can be identified in the data more precisely than the correlations of Figure 1 is another promising avenue for future research.

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A Proofs of the Main Results

Proof of Proposition 1. The proof is discussed informally in the main text. $\square$

Proof of Proposition 2. The proof proceeds in three steps.

- *Step 1: The strategy profile σ^e is an extractive social contract.*

This proof (and all the subsequent ones) make use of the one-shot deviation principle, which applies to this game because discounted payoffs are finite.

Consider first the citizen's problem. Under σ^e , her continuation value is $\bar{u}$ in every period, so the potential gain from exerting effort is $-c + \delta [\bar{u} - \bar{u}] = -c < 0$. Then, fixing the strategy of the ruler in period t , and the prescribed continuation play from $t+1$ onwards, the citizen has no incentives to exert effort in period t . Consider now her "participation choice." Fix a given tax rate τ . If $\tau \leq \tau^e(A)$, staying in the game yields at least $(1 - \tau)kA + \delta \bar{u} \geq \phi + \delta \bar{u} = \bar{u}$. Then, staying is (weakly) optimal. If $\tau > \tau^e(A)$, staying yields strictly less than $\bar{u}$, and quitting the game is optimal.

Consider now the ruler's problem. As shown above, under the prescribed continuation, the citizen never exerts effort. Therefore, the tax rate chosen by the ruler in period t has no consequences on the level of productivity in period $t+1$, so that his problem is essentially static. Hence, $\tau^e(A)$ is optimal as it maximizes his current period payoff, subject to the binding participation constraint of the citizen.

I now compute players' value functions. The citizen's value function is equal to $\bar{u}$ always. Then, the contract is always extractive on the equilibrium path. The ruler collects $\tau^e(A)kA = kA - \phi$ in taxes in every period. The growth rate of the economy is g_0 . The ruler's value function is then equal to:

$$V^e(A) = \sum_{t \geq 0} \delta^t (kA g_0^t - \phi) = \frac{kA}{1 - \delta g_0} - \frac{\phi}{1 - \delta} = \frac{kA}{1 - \delta g_0} - \bar{u},$$

where the sum converges because $\delta g_0 < \delta g_1 < 1$.

- *Step 2: The citizen's lower bound.* In any period, the citizen can always quit the game and secure the outside option. Therefore, in any social contract, the citizen's value function must (weakly) exceed $\bar{u}$.
- *Step 3: The ruler's lower bound.* Fix any social contract, σ . Suppose σ prescribes the ruler to play τ_σ in a given period. Consider a deviation in which the ruler plays some $\tau' \neq \tau_\sigma$ off path. Suppose that starting the next period, the extractive social contract is in place. By the argument above, the best deviation is $\tau' = \tau^e(A)$. The citizen's best responds by staying in the game without exerting effort. As already shown, this is a social contract. Then, the deviation is not ruled out by subgame perfection. The value of the deviation for the ruler is $V^e(A)$. Then, the ruler's continuation value along any social contract σ must be at least as large.

□

Proof of Proposition 3. Suppose strategies are Markovian and depend only on the state, A . The tax rate implemented by the ruler in period t affects neither the transition of the state nor the citizen's optimal choice of effort (which would only depend on the $t+1$ tax rate). The ruler's problem in each period is static. Therefore, he raises the tax rate up to the point at which the citizen is indifferent between quitting the game or not. It follows that the citizen's continuation value equals $\bar{u}$ at every on path history. □

Proof of Proposition 4. Road-map. The proof proceeds in two parts.

Part 1 defines and characterizes the maximum equilibrium surplus, $\bar{\Pi}(A)$ (Definition 3). I first show that first best contracts prescribe a phase of "rest" when the citizen does not exert effort, followed by a phase where the citizen exerts effort in every period (Lemma 1). Second, I turn my attention to second best contracts. In this sense, I start by deriving a productivity threshold, $\hat{A}$, above which the citizen's effort is socially desirable. I further show that effort is incentive compatible only if the citizen is granted a continuation value of at least ω (see Section 4), which the ruler can afford only above a second threshold, $\tilde{A}$. Combining the two thresholds into $A^* := \max\{\hat{A}, \tilde{A}\}$, Lemma 2 shows that second best contracts prescribe the citizen's effort if and only if $A \geq A^*$ (a

'threshold rule'). Part 1 ends by defining the calendar period T^* at which an economy growing at rate g_0 reaches the threshold A^* .

Part 2 shows that any renegotiation proof contract is also second best. I first construct a family of contracts that share the surplus $\bar{\Pi}(A)$ between the two players (Lemma 3). Whenever a contract wastes surplus, one of these contracts makes *both* players strictly better off. Since renegotiation proofness rules this out, any RP social contract must attain $\bar{\Pi}(A)$ at every history (Lemma 4). This further establishes that RP social contracts exist (Lemma 5).

The two parts are then combined to prove the three statements in Proposition 4. The proofs of all the "auxiliary" Lemmas are presented in Appendix B.

Part 1: The Maximum Equilibrium Surplus. I begin by formally defining the maximum equilibrium surplus.

Definition 3 (Maximum Equilibrium Surplus). *For any state A , let*

$$\bar{\Pi}(A) := \sup \left\{ W(\sigma|H) + V(\sigma|H) : \sigma \text{ is a social contract, and } A(H) = A \right\}.$$

The function $\bar{\Pi}(A)$ measures the largest total surplus that self-enforcing play can generate when the level of productivity is A . In this sense, $\bar{\Pi}(A)$ is a second best measure. For instance, a social planner could dictate the citizen to exert effort (and thus grow the pie), even when this is not incentive compatible for her.

In this model, taxes are ultimately transfers between the two players. It follows that the dynamics of surplus depends only on the equilibrium path of effort, provided that effort can ever be incentive compatible for the citizen.

Consider first the "auxiliary problem" of choosing a sequence of effort choices to maximize society's surplus, ignoring incentive compatibility constraints for a moment. In this regard:

Lemma 1 (Growth Never Stops). *In the auxiliary problem, if exerting effort is optimal at state A , it is optimal at every state $A' \geq A$.*

It is therefore without loss to compare only two plans, whose surpluses are $\Pi_0(A)$ and $\Pi_1(A)$, as defined in (5). Note that $V^e(A) = \Pi_0(A) - \bar{u}$. The lower bound established in Proposition 2 on the ruler's equilibrium payoff equals the present value of the slow growth path net of the citizen's outside option. In other words, since the ruler can always leave the citizen indifferent between staying in the game or not, he can always extract all the remaining resources from the slow-growing economy.

Moreover,

$$\Delta(A) > 0 \quad \Longleftrightarrow \quad A > \frac{c(1-\delta g_1)(1-\delta g_0)}{\delta(1-\delta)k(g_1-g_0)} := A^\dagger.$$

When the level of productivity is high enough ($A > A^\dagger$), the citizen's (permanent) effort can generate positive social surplus.

However, even if effort creates positive surplus, it need not be socially optimal. By Lemma 1, the only remaining question is *when* effort should start. Indeed, society can always postpone it by one period. Postponement is suboptimal when:

$$\delta[\Pi_1(Ag_1) - \Pi_1(Ag_0)] - c = \frac{\delta k A (g_1 - g_0)}{1 - \delta g_1} - c \geq 0 \quad \Longleftrightarrow \quad A \geq \hat{A} := \frac{c(1-\delta g_1)}{\delta k (g_1 - g_0)}.$$

Coherently, $\hat{A} > A^\dagger$.

Being socially desirable, however, does not make effort incentive compatible, because the citizen privately bears the cost of effort. To characterize when that is the case, I intersect three constraints. Suppose the citizen exerts effort at state A . Next period, the state is Ag_1 , and, by Lemma 1, effort continues in every subsequent period, so the continuation surplus is $\Pi_1(Ag_1)$. The first constraint is that the citizen must receive at least ω (Section 4). The second constraint is that the ruler must receive at least $V^e(Ag_1)$ (Proposition 2). Finally, feasibility requires $V^e(Ag_1) + \omega \leq \Pi_1(Ag_1)$. Then,

$$\omega + V^e(Ag_1) \leq \Pi_1(Ag_1).$$

Using $\omega = \bar{u} + \frac{c}{\delta}$ and $V^e(A) = \Pi_0(A) - \bar{u}$, the latter condition reduces to:

$$\Delta(Ag_1) \geq \frac{c}{\delta} \quad \Longleftrightarrow \quad A \geq \frac{c(1 - \delta g_1)(1 - \delta g_0)}{\delta^2(1 - \delta)k g_1(g_1 - g_0)} := \tilde{A}.$$

In a socially efficient social contract, effort must be both worthwhile and rewardable. Accordingly, define:

$$A^* := \max \left\{ \hat{A}, \tilde{A} \right\}.$$

The next result characterizes $\bar{\Pi}(A)$.¹²

Lemma 2 (Characterization of Second Best Contracts). *The maximum equilibrium surplus is:*

$$\bar{\Pi}(A) = \begin{cases} \Pi_1(A) & \text{if } A \geq A^*, \\ \sum_{s=0}^{n(A)-1} \delta^s k A g_0^s + \delta^{n(A)} \Pi_1(A g_0^{n(A)}) & \text{if } A < A^*, \end{cases} \quad (11)$$

where $n(A) := \min \{n : A g_0^n \geq A^*\}$ is the number of periods until the state reaches the threshold.

Finally, let:

$$T^* := \min \{t : A_1 g_0^{t-1} \geq A^*\} \geq 2,$$

denote the calendar time at which an economy growing at rate g_0 reaches the productivity threshold A^* , given the initial condition A_1 .

Part 2: Renegotiation Proofness and Social Welfare. The next Lemma shows that every division of the maximum equilibrium surplus which respects the lower bounds on players' payoffs (Proposition 2) is attainable in a social contract. In other words, every point on the Pareto frontier of the equilibrium payoff set is implementable as a social contract.

Lemma 3 (The Pareto Frontier is Attainable). *Fix any state $A > 0$, and any*

$$w \in [\bar{u}, \bar{\Pi}(A) - V^e(A)].$$

¹²I assume $A_1 < A^*$.

There exists a social contract, σ_w , such that, at any history H with $A(H) = A$:

$$W(\sigma_w|H) = w, \quad V(\sigma_w|H) = \bar{\Pi}(A) - w.$$

Therefore, any RP social contract is also second best from a social welfare perspective:

Lemma 4 (RP Social Contracts are Second Best). *Fix any state $A > 0$, and let $\sigma \in \Sigma(A)$. Then, at every (on or off path) history H :*

$$W(\sigma|H) + V(\sigma|H) = \bar{\Pi}(A(H)).$$

The converse of Lemma 4 also holds, and it delivers the existence of RP social contracts:

Lemma 5 (Existence of RP Social Contracts). *For every state $A > 0$ and every $w \in [\bar{u}, \bar{\Pi}(A) - V^e(A)]$, the contract σ_w constructed in the proof of Lemma 3 is renegotiation proof: $\sigma_w \in \Sigma(A)$. Therefore, $\Sigma(A) \neq \emptyset$ for all A .*

I can now prove the statements of Proposition 4, in turn.

Statement (i). By Lemma 4, σ attains $\bar{\Pi}(A(H))$ at every on path history, so it prescribes the ‘threshold rule’ of Lemma 2.

Statement (ii). Consider first the floors. The floor $\bar{u}$ follows directly from Proposition 2. Moreover, every period $t > T^*$ follows a period of effort (Statement (i)), so

$$W(\sigma|H_t) \geq \omega \tag{12}$$

by the argument presented in Section 4.

For the ceilings, I use the ruler’s lower bound. By Lemma 4, $W(\sigma|H_t) + V(\sigma|H_t) = \bar{\Pi}(A_t)$, and, by Proposition 2, $V(\sigma|H_t) \geq V^e(A_t) = \Pi_0(A_t) - \bar{u}$. Therefore:

$$W(\sigma|H_t) \leq \bar{u} + [\bar{\Pi}(A_t) - \Pi_0(A_t)].$$

If $t > T^*$, then $A_t \geq A^*$, so that $\bar{\Pi}(A_t) = \Pi_1(A_t)$ and the term in bracket is equal to $\Delta(A_t)$.

If $t \leq T^*$, then $n(A_t) = T^* - t$, and both plans prescribe no effort for $n(A_t)$ periods. Their first $n(A_t)$ terms therefore coincide, and cancel:

$$\bar{\Pi}(A_t) - \Pi_0(A_t) = \delta^{T^*-t} [\Pi_1(A_{T^*}) - \Pi_0(A_{T^*})] = \delta^{T^*-t} \Delta(A_{T^*}) .$$

This proves (9). Finally, (10) is simply the citizen's Bellman equation.

Statement (iii). By (12) and the definition of ω , at every $t > T^*$: $W(\sigma|H_t) \geq \omega = \bar{u} + \frac{c}{\delta} > \bar{u}$. Then, the contract is non-extractive at every on path history with $t > T^*$. Finally, iterating (10):

$$\sum_{s \geq 0} \delta^s (1 - \tau_{t+s}) k A_{t+s} = W(\sigma|H_t) + \frac{c}{1 - \delta} > \bar{u} = \sum_{s \geq 0} \delta^s (1 - \tau^e(A_{t+s})) k A_{t+s} .$$

Hence, $\tau_{t+s} < \tau^e(A_{t+s})$ at some $s \geq 0$. This completes the proof.

□

B Proofs of Auxiliary Lemmas

This appendix collects the proofs of the five auxiliary Lemmas stated in the Proof of Proposition 4 (Appendix A).

Proof of Lemma 1. For any plan $\{a_t\}$, society's surplus can be written as:

$$\sum_{t \geq 0} \delta^t kA \prod_{s < t} g_{a_s} - \sum_{t \geq 0} \delta^t c \mathbf{1}\{a_t = 1\} = kA \cdot G - C, \quad (13)$$

where $G := \sum_{t \geq 0} \delta^t \prod_{s < t} g_{a_s}$ and $C := \sum_{t \geq 0} \delta^t c \mathbf{1}\{a_t = 1\}$ depend on the plan but not on the state.

The value function of the auxiliary problem (say $F(A)$), is the supremum of the family of linear increasing functions defined by (13). Therefore, it is increasing and convex. Moreover, it solves the following Bellman equation:

$$F(A) = kA + \max \{ \delta F(Ag_0), -c + \delta F(Ag_1) \}.$$

Effort is optimal at state A if and only if:

$$\delta [F(Ag_1) - F(Ag_0)] \geq c. \quad (14)$$

By convexity of F , if (14) holds at A , it holds at every $A' \geq A$. □

Proof of Lemma 2. The proof follows in three steps.

- *Step 1: There is a 'threshold rule'.* By Lemma 1, surplus-maximizing effort plans imply that the citizen does not exert effort until some threshold state is reached, and exerts effort forever thereafter ($A \geq \hat{A}$). Incentive compatibility adds the constraint: $A \geq \tilde{A}$. Hence, I am left to show that a contract that prescribes effort when both constraints are satisfied ($A \geq A^*$) exists.
- *Step 2: The threshold rule can be implemented in a social contract.* Consider the following profile of strategies. The citizen exerts effort if and only if $A \geq A^*$ (and

the ruler has not deviated in the current period). She stays in the game if and only if $\tau \leq \tau^e(A)$. The ruler sets $\tau = \tau^e(A)$ up to and including the first effort period, and $\tau = 1 - \frac{\phi+c/\delta}{kA}$ afterwards. Any deviation triggers permanent reversion to σ^e , which is a social contract by Proposition 2.

The citizen's continuation value is $\bar{u}$ at every history up to and including the first effort period, and ω afterwards. Then, she has no strictly profitable deviation. Consider now the ruler. At any history, his most profitable deviation is to raise the tax to $\tau^e(A)$. Under the prescribed on path continuation, any higher rate triggers the citizen's exit. This deviation (which can possibly be profitable only at $A \geq A^*$) would allow the ruler to collect $kA - \phi$ in taxes today. As in Step 3 of the Proof of Proposition 2, this is worth

$$kA - \phi + \delta V^e(Ag_0) = V^e(A) .$$

Therefore, the ruler does not have a strictly profitable deviation if his value function (weakly) exceeds $V^e(A)$ at every on path history.

I distinguish two cases. First, at any $A \geq A^*$, his value function is equal to the total surplus net of the citizen's value function. Then, using $V^e(A) = \Pi_0(A) - \bar{u}$, the constraint reduces to:

$$\underbrace{\Pi_1(A) - \bar{u} \geq V^e(A)}_{\text{At the first effort period}} \iff \Delta(A) \geq 0 ; \quad \underbrace{\Pi_1(A) - \omega \geq V^e(A)}_{\text{At later effort periods}} \iff \Delta(A) \geq \frac{c}{\delta} ,$$

The first condition holds at any $A \geq A^*$, since $A^* \geq \tilde{A} > A^\dagger$. At any subsequent period of effort, $A \geq A^*g_1 \geq \tilde{A}g_1$, so that $\Delta(A) \geq \Delta(\tilde{A}g_1) = \frac{c}{\delta}$, by definition of $\tilde{A}$. Then, the second condition holds.

Second, at any $A < A^*$, the ruler anticipates the future stream of payoffs which will start materializing after $n(A)$ periods. Therefore, his value function is (weakly) larger than the lower bound $V^e(A)$.

- *Step 3: Computing the second best surplus.* By Step 1, no social contract generates

more surplus than a plan which could be informally described as: "Wait until the state reaches A^* , then the citizen exerts effort forever thereafter." By Step 2, this plan can be implemented as a social contract. Hence $\bar{\Pi}(A)$ equals the present value of this plan.

□

Proof of Lemma 3. The proof proceeds in three steps. Step 1 constructs two *extreme* contracts: the *ruler-preferred* contract (σ_R^*) , and the *citizen-preferred* contract (σ_C^*) . Step 2 verifies that both profiles are social contracts. Step 3 constructs the "intermediate" contracts, σ_w , by mixing over the tax rates of the two extreme contracts.

- *Step 1: The two extreme contracts.* Both contracts prescribe the citizen to exert effort if and only if $A \geq A^*$, and to stay in the game if and only if $\tau \leq \tau^e(A)$. They differ only in how taxes split the surplus generated by growth.

Moreover, the two contracts serve as each other's punishments. If the ruler deviates from the prescribed tax rate (which is different depending on the contract), play follows σ_C^* from the next period onwards. If instead the citizen deviates, play follows σ_R^* from the next period onwards. The citizen deviates when she does not exert effort at a history where effort is prescribed, when she exerts effort in a period in which the ruler deviated, or when she stays in the game after a tax rate above $\tau^e(A)$.

The ruler-preferred contract. Under σ_R^* , the citizen's value function is equal to its lower bound (which depends on whether she exerts effort or not; see Proposition 2 and Section 4), and the ruler receives the remainder of the surplus:

$$W(\sigma_R^*|H_t) = \begin{cases} \bar{u} & \text{for } t \leq T^* , \\ \omega & \text{for } t > T^* , \end{cases} \quad V(\sigma_R^*|H_t) = \bar{\Pi}(A_t) - W(\sigma_R^*|H_t) .$$

The equilibrium tax rates follow. Up to (and including) the period of the transition from stagnation to growth, the citizen's value function is constant at $\bar{u}$, so her flow

payoff is ϕ and the tax rate is exactly $\tau^e(A)$. After the transition, her value function is equal to the constant ω , so her flow payoff is the constant $(1 - \delta)\omega + c = \phi + c/\delta$. The ruler-preferred contract, σ_R^* , therefore prescribes:¹³

$$\tau_t = \tau^e(A_t) \quad \text{for } t \leq T^*, \quad \tau_t = 1 - \frac{\phi + c/\delta}{kA_t} \quad \text{for } t > T^* .$$

The citizen-preferred contract. Under σ_C^* , the ruler's value function is equal to its lower bound (Proposition 2), and the citizen receives the remainder of the surplus:

$$V(\sigma_C^*|H_t) = V^e(A_t) , \quad W(\sigma_C^*|H_t) = \bar{\Pi}(A_t) - V^e(A_t) .$$

The equilibrium tax rates follow. Before the transition ($A < A^*$), the citizen's value function is: $\bar{u} + \delta^n \Delta(A_{T^*})$, where n is the number of periods before T^* . Her flow payoff is then $(\bar{u} + \delta^n \Delta) - \delta(\bar{u} + \delta^{n-1} \Delta) = \phi$. After the transition ($A \geq A^*$), the citizen's flow payoff is $[\bar{u} + \Delta(A)] + c - \delta[\bar{u} + \Delta(A_{g_1})] = \phi + \frac{\delta k(g_1 - g_0)}{1 - \delta g_0} A$, using the fact that Δ is linear in the state. The citizen-preferred contract, σ_C^* , therefore prescribes:

$$\tau_t = \tau^e(A_t) \quad \text{for } t < T^*, \quad \tau_t = \tau^e(A_t) - \frac{\delta(g_1 - g_0)}{1 - \delta g_0} \quad \text{for } t \geq T^* .$$

- *Step 2: Both profiles are social contracts.* Because each contract serves as the other's punishment, their equilibrium properties are verified jointly.

- *The citizen has no strictly profitable deviations.* Consider first her choice of effort, at a history where effort is prescribed by the candidate social contract. The current period's flow payoff is the same regardless on the effort choice. Complying with the contract costs c today, and leads to either σ_R^* or σ_C^* at state Ag_1 , which implies a continuation value of at least ω . Shirking costs nothing today, but leads to a "restarted" σ_R^* contract, at state Ag_0 , which

¹³ Assuming that $kA^* \left(1 - \frac{\delta(g_1 - g_0)}{1 - \delta g_0}\right) \geq \phi$ and $kA^* g_1 \geq \phi + c/\delta$, keeps both contracts' tax rates in $[0, 1]$ at every reachable state.

is worth $\bar{u}$.¹⁴ Then, complying to the contract is always (weakly) optimal. Finally, since both contracts leave the citizen with a value function of at least $\bar{u}$ in every period, she does not want to quit the game.

- *The ruler has no strictly profitable deviation.* At any history, the ruler's most profitable deviation is to raise the tax rate to $\tau^e(A)$. Given the prescribed play, any higher rate triggers the citizen's exit.¹⁵ This deviation would allow the ruler to collect $kA - \phi$ in taxes today.

How would the citizen respond to such deviation? She never finds it (strictly) optimal to leave the game. Consider her choice of effort.¹⁶ Exerting effort in a period in which the ruler deviated is itself a deviation, and therefore leads to σ_R^* , which yields $-c + \delta\omega = \delta\bar{u}$. By contrast, withdrawing effort leads to σ_C^* , and yields $\delta[\bar{u} + \bar{\Pi}(Ag_0) - \Pi_0(Ag_0)] \geq \delta\bar{u}$. Then, the ruler's deviation forces the citizen to withdraw effort for one period, followed by σ_C^* forever thereafter. For the ruler, this is worth

$$kA - \phi + \delta V^e(Ag_0) = V^e(A) .$$

Then, it is never (strictly) profitable.

- *Step 3: The intermediate contracts.* Fix some $w \in [\bar{u}, \bar{\Pi}(A) - V^e(A)]$. There exists $\lambda \in [0, 1]$ such that $w = \lambda [\bar{\Pi}(A) - V^e(A)] + (1 - \lambda) \bar{u}$.

The candidate contract, σ_w , prescribes the same actions and the same punishments as the two extreme contracts. The citizen exerts effort if and only if $A \geq A^*$, stays in the game if and only if $\tau \leq \tau^e(A)$, and rests in any period in which the ruler

¹⁴The two "branches" of W in Step 1 are indexed by calendar time because, on path, effort begins at T^* . However, what σ_R^* actually tracks is whether effort has been exerted by the citizen in the previous period. Therefore, a punishment "restarts" σ_R^* , at a phase in which the citizen had exerted no effort, so it is worth $\bar{u}$ at any state — including states reached after T^* . The same applies to the citizen's deviations under σ_C^* , considered next.

¹⁵The citizen's exit is sequentially rational at that history. Both players have deviated from the prescribed contract (the ruler by implementing a higher tax rate, the citizen by staying in the game nonetheless), so suppose that play follows σ_R^* from the next period onwards. This is worth $\bar{u}$ to the citizen at every state. Staying therefore yields $(1 - \tau)kA + \delta\bar{u} < \phi + \delta\bar{u} = \bar{u}$.

¹⁶This response matters because it determines how fast the state grows in the deviation period, and hence how much the deviation is worth to the ruler.

deviated. On path, in every period, the ruler sets the λ -weighted average of the tax rates prescribed by σ_C^* and σ_R^* . Off path, if the ruler deviates, play follows σ_C^* ; if the citizen deviates, play follows σ_R^* .

Three observations complete the proof.

First, *feasibility*. In every period, the prescribed tax rate is a convex combination of the two extreme rates. Under the maintained parametric assumptions (see Footnote 13), both lie in $[0, \tau^e(A)]$; hence, so does their average.

Second, *second best efficiency*. On path, the three contracts prescribe identical actions, and differ only in the equilibrium tax rates, which are pure transfers. Since the citizen's flow payoff is linear in the tax rate, at every on path history her continuation value under σ_w is the λ -average of her values under σ_C^* and σ_R^* . Since total surplus is always maximized and equal to $\bar{\Pi}(A)$, the ruler receives the remainder. In particular, the citizen's value at the initial state is w , and the ruler's is $\bar{\Pi}(A) - w$.

Third, *incentive compatibility*. The checks on players' incentive compatibility constraints follow directly from those in Step 2.

□

Proof of Lemma 4. Fix $\sigma \in \Sigma(A)$ and a history H . Let $w := W(\sigma|H)$ and $v := V(\sigma|H)$. By Definition 3, $w + v \leq \bar{\Pi}(A(H))$.

Suppose, towards a contradiction, that $w + v < \bar{\Pi}(A(H))$. Pick any value w' strictly between w and $\bar{\Pi}(A(H)) - v$. By Proposition 2, $w \geq \bar{u}$ and $v \geq V^e(A(H))$. Therefore:

$$w' \in (w, \bar{\Pi}(A(H)) - v) \subseteq [\bar{u}, \bar{\Pi}(A(H)) - V^e(A(H))] .$$

At the beginning of every period, the continuation game is identical to the game with initial state $A(H)$. Then, by Lemma 3, there exists a social contract of the continuation game, σ' , such that:

$$W(\sigma'|H) = w' > w \quad \text{and} \quad V(\sigma'|H) = \bar{\Pi}(A(H)) - w' > v .$$

The contract σ' makes both players strictly better off, contradicting Definition 2. $\square$

Proof of Lemma 5. Fix any $w \in [\bar{u}, \bar{\Pi}(A) - V^e(A)]$. At every on path history of σ_w , players' value functions sum to $\bar{\Pi}(A(H))$ (Step 3 of the proof of Lemma 3). At every off path history, continuation play is a restarted σ_R^* or σ_C^* , whose value functions also sum to $\bar{\Pi}(A(H))$ (Step 1). By Definition 3, any social contract σ' of the continuation game satisfies $W(\sigma'|H) + V(\sigma'|H) \leq \bar{\Pi}(A(H))$. Hence, at no history can an alternative social contract make *both* players strictly better off. Then, by Definition 2: $\sigma_w \in \Sigma(A)$. $\square$

C Data Appendix: Construction of Figure 1

This appendix explains the construction of Figure 1.

Institutions The measure of constraints on rulers is the "legislative constraints on the executive" index of the V-Dem project, version 15 (variable `v2xlg_legcon`), often referred to as a *checks and balances* index. It ranges from 0 to 1 and aggregates expert codings of whether the legislature and government agencies can question, investigate, and oversee the executive. Panel (a) plots its unweighted mean across all polities observed in a given year, 1789–2024.

Technology Technology adoption at the country-year level is measured from the CHAT dataset (Comin and Hobijn [2009]). To obtain the plotted measure, I perform the following steps. I first drop World Bank regional aggregates to obtain one observation for each polity-year. Then, I drop duplicated variables that measure the same technology (e.g., the same technology recorded in different ways), obtaining 78 *unique* technologies. A country is coded as having adopted a technology by year t if any of its source variables records a strictly positive value in some year up to t . Finally, each technology enters the yearly world frontier in the first year in which *any* country records its adoption. Panel (b) plots the size of the frontier over 1750–2023.

The adoption index and Panel (c) A country-year's *technology adoption index* is the share of the world frontier the country has adopted. I merge the index with the checks and balances index on country and year, and pool all country-years between 1850 and 2000, ending up with $N = 14,707$ observations from 175 countries.

For Panel (c), observations are sorted by the adoption index and split into 40 equal-sized bins (≈ 368 country-years each); each dot plots the within-bin means of both variables. The fitted line is estimated by OLS on the underlying, unbinned, observations: slope = 0.453 (intercept = 0.267), $r = 0.424$, $t = 56.7$, $F(1, 14,705) = 3,215$ ($p < 0.001$). Note that these are pooled correlations, mixing cross-country and within-country variation.